

## **The problem with the OME metric**

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# Abstract

**i** Note

[1] "Low quality document, \n edit the index.qmd to ` PRODUCTION\_QUALITY <- TRUE ` before r

Abstract to be written

# 1 Manuscript

## 1.1 Origin of opioid equivalence and equipotency

Opioid medications are an important pillar of analgesia in medicine, used across a range of specialities and patients. There are a wide range of opioids, some naturally derived with others synthetic, each with their own potency. Whilst a wide range of factors may influence the potency of a specific opioid in a specific patient, there has long been a need to calculate the equi-analgesic dose, or the dose of a medication that would provide an equal amount of analgesia compared to a reference.

The need for agreed conversion values has lead to a number of publications over the last 20 years aiming to formalise the differing potencies. Much of this literature comes from the domain of palliative care in the context of opioid rotation and looked at multiple single opioid small studies to establish conversion values. McPherson described a range of limitations in the use of these equianalgesic tables including the source of the data, lack of patient specific variables and bi-directionality of equivalence values. [1]. Despite this, a set of approximate conversion values, taken with clinical experience offer a helpful set of starting points when prescribing different opioids

The equi-analgesic conversion set have offered a new opportunity in research, the ability to quantify the total opioid burden for a group of patients. With the increase in electronic health records, opportunities to now use total opioid use as either a feature or an outcome has become increasingly prevalent. In the last 5 years there have been 4,031 papers using the terms “*Milligram morphine equivalent*” (MME) or “*Oral morphine equivalent*” (OME) (see Section ??).

In 2016 Neilsen et al published on the first formal OME conversion tables for the intention of research [2]. The authors aimed to source a range of national guidelines and recommended a single value for each opioid and route with a upper and lower quartile. This “common conversion set” paved the way for the idea that all opioid research should be using a single unified conversion set. It was clear in the literature that using a range of different conversion sets would not produce equitable or comparable results. To the extent that in 2022 the CDC produced their own value set an obligated any NIH researches working in opioid research to use their opioid conversions [3].

A range of papers have highlightes some of the limitations and issues with opioid conversions sets in research. Namely the lack of accounting for inter-patient variability and the differential applicaiton of conversion sets. [Add citations]

Two aspects that have not been covered are the dependance on the proportion of each opioid and the error within the conversion values used. [Add citations about who said/didnt say what]

## 1.2 Calculating opioid equivalence in clinical practise

Often in clinical practice the term potency is used to describe the relative strength of one opioid compared to another. The equi-potency of a medication is commonly understood to be the dose required to achieve equal effect, and in the case of opioids, the equianalgesic effect. Thus Equation ?? describes the relationship between the conversion value  $c$  and the potency  $\rho$  of a given medication  $i$  via a given route  $r$ .

$$c_{i,r} = \frac{1}{\rho_{i,r}} \quad (1.1)$$

The value of  $c$ , then allows for the calculation of the equivalent effective dose of a target medication  $d_{\text{target}}$ , given a dose of a reference medication  $d_{\text{reference}}$  (Equation ??).

$$d_{\text{reference}} \times c_{i,r} = d_{\text{target}} \quad (1.2)$$

A full worked example of this is process of opioid conversion calculation is provided in Section ??.<sup>1</sup>

## 1.3 Calculating opioid equivalence in research

The technique of comparing opioid equivalent doses has lead to the development of a new metric, the sum opioid burden. By converting each medication into its equi-analgesic dose of the reference medication (almost exclusivly oral morphine) and then calculating the sum, one could estimate the total opioid analgesic effect administered.

The opioid burden metric is often presented as the weighted sum of opioids administered. We can define  $\text{OME}_j$  as the sum of the dose  $d$  for medication  $i$  in the set of medications  $\mathbf{M}$

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<sup>1</sup>This approach is used clinically for opioid conversion or rotation and is the origin of the opioid equivalence system. With a small set of opioid equivalence values, it's possible to convert doses between a wide range of different opioids, often via the reference medication as an intermediate.